Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The vapour density of undecomposed N_2O_4 is 46. When heated, vapour density decreases to 24.5 due to its dissociation to NO_2. The % dissociation of N_2O_4 at the final temperature is

(A) 80

(B) 60

(C) 40

(D) 70

Q.2

If $CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(l)}$
$K_p = 1.086 \times 10^{-4}\ atm^2$ at $25^\circ C$. The efflorescent nature of $CuSO_4.5H_2O$ can be noticed when vapour pressure of H_2O in atmosphere is

(A) > 7.29 mm

(B) < 7.92 mm

(C) ≥ 7.92 mm

(D) None

Q.3

The enthalpies of two reaction are ΔH_1 and ΔH_2 (both positive) with $\Delta H_2 > \Delta H_1$. If the temperature of reacting system is increased from T_1 to T_2, predict which of the following alternatives is correct?

(A) $\frac{K_I'}{K_1} = \frac{K_2'}{K_2}$

(B) $\frac{K_1'}{K_1} > \frac{K_2'}{K_2}$

(C) $\frac{K_1'}{K_L} < \frac{K_2'}{K_2}$

(D) None

Q.4

Gaseous N_2O_4 dissociates into gaseous NO_2 according to the reaction

$$[N_2O_4(g) \rightleftharpoons 2NO_2(g)]$$

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At $300\ \mathrm{K}$ and 1 atm pressure, the degree of dissociation of $\mathrm{N_2O_4}$ is 0.2. If one mole of $\mathrm{N_2O_4}$ gas is contained in a vessel, then the density of the equilibrium mixture is :

(A) $1.56\ \mathrm{g/L}$

(B) $6.22\ \mathrm{g/L}$

(C) $3.11\ \mathrm{g/L}$

(D) $4.56\ \mathrm{g/L}$

Q.5

The value of K_p for the equilibrium reaction $\mathrm{N_2O_4}(g) \rightleftharpoons 2\mathrm{NO_2}(g)$ is 2
The percentage dissociation of $\mathrm{N_2O_4}(g)$ at a pressure of $0.5\ \mathrm{atm}$ is

(A) 25

(B) 88

(C) 50

(D) 71

Q.6

At $527^{\circ}\mathrm{C}$, the reaction given below has $\mathrm{K_c} = 4$
$\mathrm{NH_3(\ g)} \rightleftharpoons \frac{1}{2}\ \mathrm{N_2(\ g)} + \frac{3}{2}\mathrm{H_2(\ g)}$
What is the $\mathrm{K_P}$ for the reaction? $\mathrm{N_2(\ g)} + 3\mathrm{H_2(\ g)} \rightleftharpoons 2\mathrm{NH_3(\ g)}$

(A) $16 \times (800\mathrm{R})^2$

(B) $\left(\frac{800\mathrm{R}}{4}\right)^{-2}$

(C) $\left(\frac{1}{4\times 800\mathrm{R}}\right)^{2}$

(D) None of these

Q.7

In reaction $\mathrm{A} + 2\ \mathrm{B} \Rightarrow 2\mathrm{C} + \mathrm{D}$, initially concentration of B was 1.5 times of $[\mathrm{A}]$, but at equilibrium the concentrations of A and B became equal. The equilibrium constant for the reaction is:

(A) 8

(B) 4

(C) 12

(D) 6

Q.8

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What happens when an inert gas is added to an equilibrium keeping volume unchanged?

(A) More product will form

(B) Less product will form

(C) More reactant will form

(D) Equilibrium will remain unchanged

Q.9

In a saturated solution of the sparingly soluble strong electrolyte $\mathrm{AgIO_3}$ (molecular mass $= 283$) the equilibrium which sets is $\mathrm{AgIO_3(\ s) \rightleftharpoons Ag^+(aq) + IO_3^-(aq)}$. If the solubility product constant K_{sp} of $\mathrm{AgIO_3}$ at a given temperature is 1.0×10^{-8}, what is the mass of $\mathrm{AgIO_3}$ contained in $100\ \mathrm{mL}$ of its saturated solution?

(A) $1.0 \times 10^{-4}\ \mathrm{g}$

(B) $28.3 \times 10^{-2}\ \mathrm{g}$

(C) $2.83 \times 10^{-3}\ \mathrm{g}$

(D) $1.0 \times 10^{-7}\ \mathrm{g}$.

Q.10

If pressure is applied to the following equilibrium, liquid $\rightleftharpoons$ vapours the boiling point of liquid

(A) will increase

(B) will decrease

(C) may increase or decrease

(D) will not change

Q.11

Which of the following will not change the concentration of ammonia at the equilibrium?

(A) Increase of pressure

(B) Increase of volume

(C) Addition of catalyst

(D) Decrease of temperature

Q.12

The exothermic formation of $\mathrm{ClF_3}$ is represented by the equation:

$\mathrm{Cl_2(\ g) + 3\ F_2(\ g) \rightleftharpoons 2ClF_3(\ g);\ \Delta H = -329\ kJ}$

Which of the following will increase the quantity of $\mathrm{ClF_3}$ in an equilibrium mixture of $\mathrm{Cl_2}$, $\mathrm{F_2}$ and $\mathrm{ClF_3}$?

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(A) Adding F_2

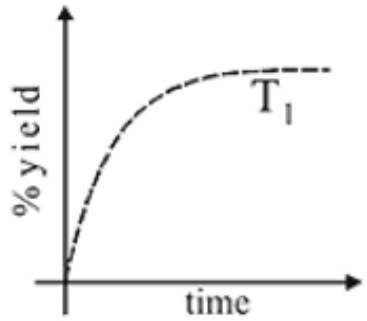
(B) Increasing the volume of the container

(C) Removing Cl_2

(D) Increasing the temperature

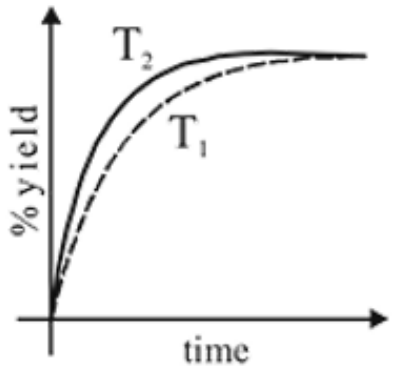
Q.13

The % yield of ammonia as a function of time in the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), \Delta H < 0$ at (P, T_1) is given below

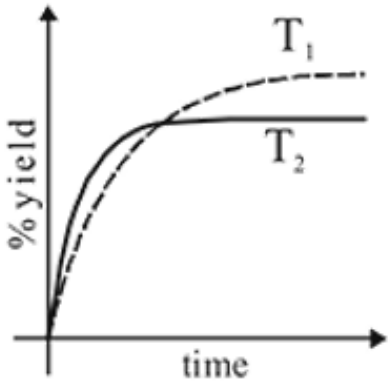


If this reaction is conducted at (P, T_2), with $T_2 > T_1$, the \%yield of ammonia as a function of time is represented by

(A)

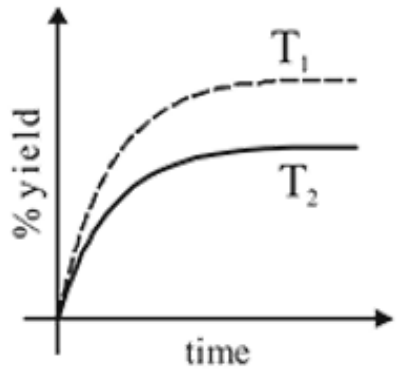


(B)

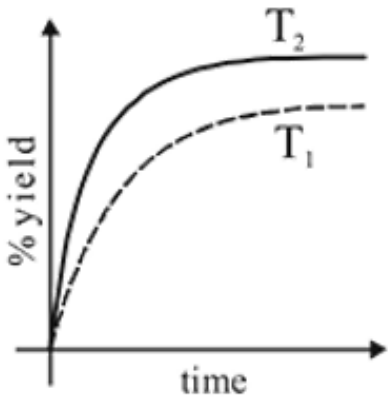


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(C)



(D)



Q.14

Some inert gas is added at constant volume to the following reaction at equilibrium $\mathrm{NH_4HS(s)} \rightleftharpoons \mathrm{NH_3(\,g) + H_2\,S(\,g)}$

Predict the effect of adding the inert gas:

(A) The equilibrium shifts in the forward direction

(B) The equilibrium shifts in the backward direction

(C) The equilibrium remains unaffected

(D) The value of $\mathrm{K_P}$ is increased

Q.15

The standard state Gibb's free energy change for the isomerisation reaction cis-2-pentene $\rightleftharpoons$ trans-2-pentene is $-3.67\ \mathrm{kJ\ mol^{-1}}$ at $400\ \mathrm{K}$. If more trans-2-pentene is added to the reaction vessel

(A) more cis-2-pentene is formed

(B) equilibrium shifts in the forward direction

(C) equilibrium remains unaltered

(D) more trans-2-pentene is produced

Q.16

(1) $\mathrm{N_2(\,g) + 3H_2(\,g) \rightleftharpoons 2NH_3(\,g), K_1}$
(2) $\mathrm{N_2(\,g) + O_2(\,g) \rightleftharpoons 2NO(g), K_2}$

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(3) $H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons = H_2O(g), K_3$

The equation for the equilibrium constant of the reaction

$2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$

(K_4) in terms of K_1, K_2 and K_3 is :

(A) $\frac{K_1 \cdot K_2}{K_3}$

(B) $\frac{K_1 \cdot K_3^2}{K_2}$

(C) $K_1\ K_2\ K_3$

(D) $\frac{K_2 \cdot K_3^3}{K_1}$

Q.17

On the basis of Le-Chatelier's principle, predict which of the following conditions would be unfavourable for the formation of SO_3 ? Given that $2SO_2 + O_2 \rightleftharpoons 2SO_3; \Delta H = -42\text{kcal}$

(A) Low pressure and low temperature

(B) High pressure and low temperature

(C) High temperature and low pressure

(D) High concentration of SO_2

Q.18

The reaction, $SO_2 + Cl_2 \longrightarrow SO_2Cl_2$ is exothermic and reversible. A mixture of $SO_2(g)$, Cl_2 (g) and $SO_2Cl_2(l)$ is at equilibrium in a closed container. Now a certain quantity of extra SO_2 is introduced into the container, the volume remaining the same. Which of the following is are true?

(A) The pressure inside the container will not change.

(B) The temperature will not change.

(C) The temperature will increase.

(D) The temperature will decrease.

Q.19

Why only As^{3+} gets precipitated as $As_2\ S_3$ and not Zn^{2+} as ZnS when $H_2\ S$ is passed through an acidic solution containing As^{3+} and Zn^{2+} ?

(A) Solubility product of $As_2\ S_3$ is less than that of ZnS

(B) Enough As^{3+} are present in acidic medium

(C) Zinc salt does not ionise in acidic medium

(D) Solubility product changes in presence of an acid

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Q.20

Hemoglobin (Hb) protein transport O_2 in blood and each Hb can bind $4O_2$, molecules. In discussing protein oxygen binding capacity, biochemists use a measure called the P^{50} value, defined as the partial pressure of oxygen at which 50% of hemoglobin is saturated. Fetal hemoglobin has a P^{50} value of 20 torr and adult has P^{50} value of 40 torr. For the following equilibrium:

$4O_2(\ g) + Hb(aq) \rightarrow HbO_{2_4}(aq)$

What is the value of K_{fetal} / K_{adult} ?

[Where K represents equilibrium constant.]

(A) 18

(B) 16

(C) 20

(D) 22

Answers & Solutions

Q.1 Answer:

80

Solution:

$N_2O_4 \rightleftharpoons 2NO_2$

1 0 at initial

$1-\alpha \quad 2\alpha$ **at equilibrium**

$\therefore \frac{V \cdot D_{initial}}{V \cdot D_{final}} = \frac{n_{final}}{n_{initial}}$

$\frac{46}{25.4} = 1.8 = 1 + \alpha$

$\alpha = 0.87$ **close to 87%**

Q.2 Answer:

< 7.92 mm

Solution:

An efflorescent salt is one that loses H_2O to atmosphere.

For the reaction

$CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(1)}$

$K_p = \left(p'_{H_2O}\right)^2 = 1.086 \times 10^{-4}$

$p'_{H_2O} = 1.042 \times 10^{-2}$ atm $= 7.92$ mm

$\because$ **If p'_{H2O} at $25^\circ C < 7.92$ mm only then, reaction will proceed in forward direction.**

Q.3 Answer:

$\frac{K'_1}{K_L} < \frac{K'_2}{K_2}$

Solution:

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As the temperature of reacting system is increased the equilibrium constant of reaction is also increased for endothermic reactions so for two reactions on increasing the temperature by equal amounts $\frac{K_1'}{K_1} < \frac{K_2'}{K_2}$

Q.4 Answer:

$3.11\ g/L$

Solution:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

$t = 0$	1	0
$t = eq.$	$1-\alpha$	2α

Where α **= Degree of dissociation. Mol. wt. of mixture**

$= \frac{(1-\alpha)\times M_{N_2O_4} + 2\alpha \times M_{NO_2}}{(1+\alpha)}$

$= \frac{(1-0.2)92 + 2\times 0.2 \times 46}{(1+0.2)} = 76.66$

Now, As per ideal gas equation $PV = nRT$

$PM_{mixture} = dRT$

$\therefore d = \frac{PM_{mix}}{RT} = \frac{1\times 76.66}{0.0821\times 300} = 3.11\ g/L$

Q.5 Answer:

71

Solution:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

Initial moles	1	0
Moles at equil.	$(1-\alpha)$	2α

(α = degree of dissociation)

Total number of moles at equil.

$= (1-\alpha) + 2\alpha$

$= (1+\alpha)$

$$p_{N_2O_4} = \frac{(1-\alpha)}{(1+\alpha)} \times P$$

$$p_{NO_2} = \frac{2\alpha}{(1+\alpha)} \times P$$

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$K_P = \frac{(p_{NO_2})^2}{p_{N_2O_4}} = \frac{\left(\frac{2\alpha}{(1+\alpha)} \times P\right)^2}{\left(\frac{1-\alpha}{1+\alpha}\right) \times P} = \frac{4\alpha^2 P}{1-\alpha^2}$

Given, $K_P = 2, P = 0.5$ atm

$$\therefore \quad K_P = \frac{4\alpha^2 P}{1-\alpha^2}$$

$$= \frac{4\alpha^2 \times 0.5}{1-\alpha^2}$$

$\alpha = 0.707 \approx 0.71$

$\therefore$ **Percentage dissociation** $= 0.71 \times 100 = 71$

Q.6 Answer:

$\left(\frac{1}{4 \times 800\text{R}}\right)^2$

Solution:

If, $\text{K}_\text{c} = \frac{[\text{N}_2]^{1/2}[\text{H}_2]^{3/2}}{[\text{NH}_3]^1} = 4$

then, $\frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = (4)^{-2} = \frac{1}{16}$

also $K_P = K_c \cdot (RT)^{\Delta n_g}$

$= \frac{1}{16} \times (800\text{R})^{-2} = \left(\frac{1}{4 \times 800\text{R}}\right)^2$

Q.7 Answer:

4

Solution:

$$\begin{array}{ccccccc} \text{A} & + & 2\text{B} & \rightleftharpoons & 2\text{C} & + & \text{D} \\ a & & 1.5a & & 0 & & 0 \\ (a-x) & & (1.5a-2x) & & 2x & & x \end{array}$$

Hence $K_C = \frac{(2x)^2 \times x}{(a-x)(1.5a-2x)^2}$

Given, at equilibrium $\therefore \quad (a-x) = (1.5a-2x)$

$\therefore \quad a = 2x$

On solving $\text{K}_\text{C} = 4$

Q.8 Answer:

Equilibrium will remain unchanged

Solution:

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On adding inert gas at constant volume the total pressure of the system is increased, but the partial pressure of each reactant and product remains the same. Hence no effect on the state of equilibrium.

Q.9 Answer:

2.83×10^{-3} g

Solution:

Let s = solubility

$AgIO_3 \rightleftharpoons Ag^+ + IO_3^-$

$K_{sp} = [Ag^+][IO_3^-] = s \times s = s^2$

Given $K_{sp} = 1 \times 10^{-8}$

$\therefore \quad s = \sqrt{K_{sp}} = \sqrt{1 \times 10^{-8}}$

$= 1.0 \times 10^{-4}$ mol/lit $= 1.0 \times 10^{-4} \times 283$ g/lit

($\because$ **Molecular mass of** $AgIO_3 = 283$)

$= \frac{1.0 \times 10^{-4} \times 283 \times 100}{1000}$ g/100 mL

$= 2.83 \times 10^{-3}$ g/100 mL

Q.10 Answer:

will increase

Solution:

Boiling point of a liquid is the temperature at which vapour pressure became equal to atm pressure. If the pressure is applied to the above equilibrium the reaction will go to the backward direction, i.e vapour pressure decrease hence the boiling point increase.

Q.11 Answer:

Addition of catalyst

Solution:

Catalyst does not disturb the equilibrium state.

Q.12 Answer:

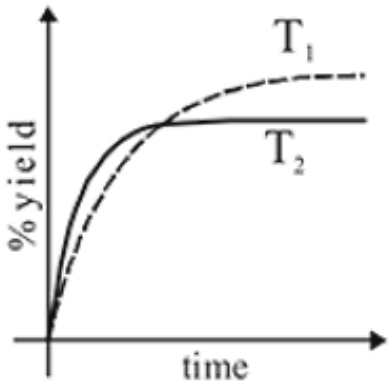
Adding F_2

Solution:

The reaction given is an exothermic reaction thus according to Le chatalier's principle lowering of temperature, addition of F_2 and Cl_2 favour the forward direction and hence the production of ClF_3.

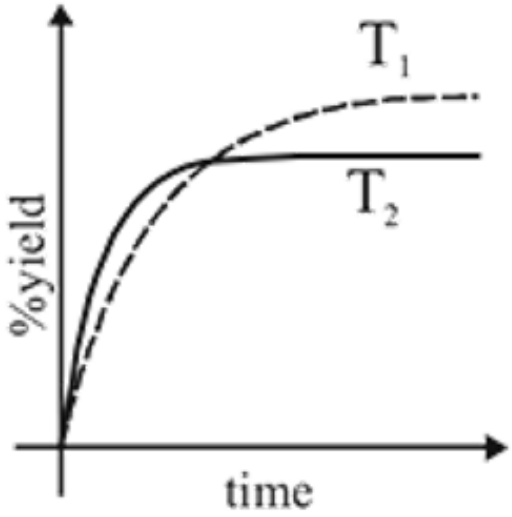
Q.13 Answer:

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Solution:

Initially on increasing temperature rate of reaction will increase, so $\%$ yield will also increase with time. But at equilibrium $\%$ yield at high temperature (T_2) would be less than at T_1 as reaction is exothermic so the graph is



Q.14 Answer:

The equilibrium remains unaffected

Solution:

Addition of inert gas at constant volume has no effect on equilibrium.

Q.15 Answer:

more cis-2-pentene is formed

Solution:

Equilibrium shifted in the backward direction. (Le Chatelier's principle)

Q.16 Answer:

$\frac{K_2 \cdot K_3^3}{K_1}$

Solution:

To calculate the value of K_4 in the given equation we should apply :

eqn. (2)+ eqn.(3)\times3-eqn. (1)

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hence $K_4 = \frac{K_2 K_3^3}{K_1}$

Q.17 Answer:

Low pressure and low temperature

Solution:

Since reaction is exothermic hence low temperature will favour forward reaction also volume is decreased by applying high pressure

Q.18 Answer:

The temperature will increase.

Solution:

By addition of SO_2 equilibrium will shift to RHS which is exothermic. Hence temperature will increase.

Q.19 Answer:

Solubility product of $As_2 S_3$ is less than that of ZnS

Solution:

K_{sp} **of** $As_2 S_3$ **is less than** ZnS**. In acidic medium ionisation of** $H_2 S$ **is suppresed (common ion effect) and** K_{sp} **of** ZnS **does not exceed.**

Q.20 Answer:

16

Solution:

$$K_{fetal} = \frac{HbO_{2_4}}{[Hb] \cdot P_{O_2}^4} = \frac{HbO_{2_4}}{[Hb] \times (20torr)^4}$$

$$K_{adult} = \frac{HbO_{2_4}}{[Hb] \times (40torr)^4}$$

$$\therefore \quad \frac{K_{fetal}}{K_{adult}} = \frac{40^4}{20} = 16$$

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